

OpenStack云平台综合部署与管理

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- 2 创建cloud-init配置文件
- 3.设计安全组最小权限策略
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- 5.验证配置
- 6 创建第二台云主机验证一致性
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一、总体要求

项目名称：OpenStack云平台全生命周期管理与网络架构实践

核心目标：通过完整的工作流，掌握OpenStack核心组件（Neutron、Nova、Glance）的协同操作，实现从网络搭建、云主机部署实践。

二、具体要求

第一部分：网络架构设计与实现

1. 网络拓扑搭建

- 使用CLI创建2个网络（如：内部网络、外部网络）
- 每个网络至少包含1个子网，使用不同CIDR段（如：192.168.10.0/24, 192.168.20.0/24）
- 创建路由器实现网络间通信

2. 端口与联通验证

- 为后续云主机创建虚拟端口
- 部署2台云主机分别位于不同子网
- 验证：同子网互通、跨子网路由、外部网络访问

3. 故障注入与排错

- 模拟故障（如：安全组误配、路由表错误）
- 记录排查过程与修复方案

第二部分：云主机生命周期管理

1. 基础操作

- CLI完成：创建/启动/停止/重启/暂停/恢复/锁定/解锁/删除

- 每种操作需截图并说明适用场景

2. 快照管理

- 创建运行中云主机快照
- 使用快照：①重建新实例 ②恢复原实例
- 对比“快照”与“镜像”的差异，给出使用建议矩阵

3. 批量脚本

- 编写Shell/Python脚本批量创建3台云主机
- 包含错误处理（资源不足、网络异常等）

第三部分：镜像初始化与安全加固

1. 密钥对管理

- 创建SSH密钥对，配置多用户访问
- 实现无密码SSH登录验证

2. cloud-init自动化

- 编写cloud-init配置完成：
 - 创建sudo用户
 - 部署公钥
 - 安装nginx/mysql等至少1个服务
 - 写入自定义配置文件
- 验证初始化结果

3. 安全组策略

- 设计最小权限安全组：
 - 仅开放必要端口（SSH:22, HTTP:80等）
 - 按源IP限制访问
- 测试策略有效性（合规访问应成功，违规访问应被拒绝）

三、执行过程

openstack 环境安装

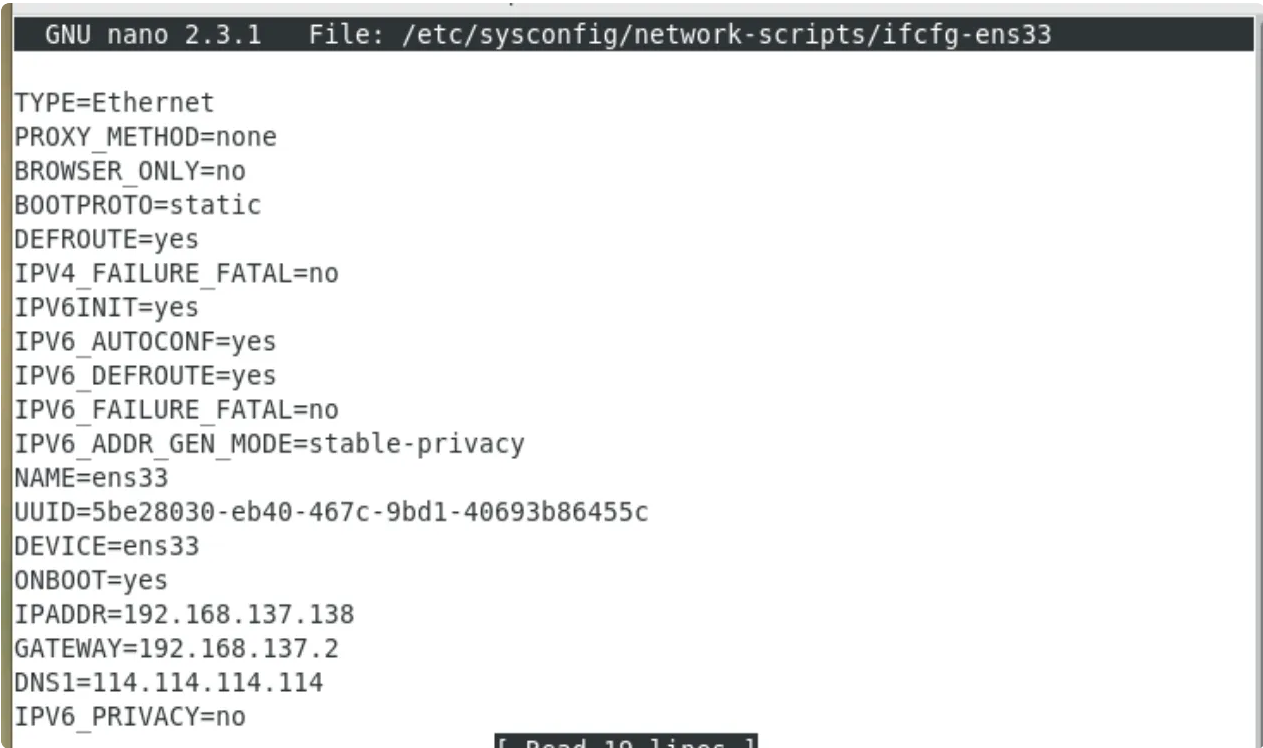
1.创建全新 centos7 虚拟机环境 使用工具 VMware

问题难点：网络的配置

使用 NAT 模式 分配 ip 地址 192.168.137.138
正常创建完后网络无法正常连接，我们需要做出一些操作

Python |

```
1 #编辑配置文件并重启网络服务
2 [root@localhost wushuyue]# nano /etc/sysconfig/network-scripts/ifcfg-ens33
3 [root@localhost wushuyue]# nano /etc/resolv.conf
4 [root@localhost wushuyue]# systemctl restart network
```



```
GNU nano 2.3.1 File: /etc/sysconfig/network-scripts/ifcfg-ens33

TYPE=Ethernet
PROXY_METHOD=none
BROWSER_ONLY=no
BOOTPROTO=static
DEFROUTE=yes
IPV4_FAILURE_FATAL=no
IPV6INIT=yes
IPV6_AUTOCONF=yes
IPV6_DEFROUTE=yes
IPV6_FAILURE_FATAL=no
IPV6_ADDR_GEN_MODE=stable-privacy
NAME=ens33
UUID=5be28030-eb40-467c-9bd1-40693b86455c
DEVICE=ens33
ONBOOT=yes
IPADDR=192.168.137.138
GATEWAY=192.168.137.2
DNS1=114.114.114.114
IPV6_PRIVACY=no
```

要点：BOOTPROTO = static ONBOOT=yes 添加 IPADDR 和 GATEWAY 增加 DNS(114.114.114.114)



```
GNU nano 2.3.1 File: /etc/resolv.conf

# Generated by NetworkManager
nameserver 114.114.114.114
```

验证网络

1. 使用 ip addr 看到 ens33 有 inet 字段


```

2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP g
roup default qlen 1000
    link/ether 00:0c:29:b9:38:59 brd ff:ff:ff:ff:ff:ff
    inet 192.168.137.138/24 brd 192.168.137.255 scope global noprefixroute ens33
        valid_lft forever preferred_lft forever
    inet6 fe80::da82:112b:4901:a16c/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: virbr0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN

```

2.ping 百度可以 ping 通

```

[root@localhost wushuyue]# ping www.baidu.com
PING www.a.shifen.com (223.109.82.212) 56(84) bytes of data.
64 bytes from 223.109.82.212 (223.109.82.212): icmp_seq=1 ttl=128 time=15.4 ms
64 bytes from 223.109.82.212 (223.109.82.212): icmp_seq=2 ttl=128 time=16.7 ms
64 bytes from 223.109.82.212 (223.109.82.212): icmp_seq=3 ttl=128 time=15.5 ms
64 bytes from 223.109.82.212 (223.109.82.212): icmp_seq=4 ttl=128 time=15.8 ms
64 bytes from 223.109.82.212 (223.109.82.212): icmp_seq=5 ttl=128 time=20.3 ms

```

2.环境安装

关闭防火墙和取消防火墙开机自启动

```

[root@localhost wushuyue]# systemctl stop firewalld
[root@localhost wushuyue]# systemctl disable firewalld
Removed symlink /etc/systemd/system/multi-user.target.wants/firewalld.service.
Removed symlink /etc/systemd/system/dbus-org.fedoraproject.FirewallD1.service.
[root@localhost wushuyue]#

```

关闭 NetworkManager服务

```

[root@localhost wushuyue]# systemctl stop NetworkManager
[root@localhost wushuyue]# systemctl disable NetworkManager
Removed symlink /etc/systemd/system/multi-user.target.wants/NetworkManager.service.
Removed symlink /etc/systemd/system/dbus-org.freedesktop.nm-dispatcher.service.
Removed symlink /etc/systemd/system/network-online.target.wants/NetworkManager-wait-online.service.

```

永远关闭 SeLinux

```

[root@localhost wushuyue]# nano /etc/selinux/config

```

```
File Edit View Search Terminal Help
GNU nano 2.3.1 File: /etc/selinux/config Modified

# This file controls the state of SELinux on the system.
# SELINUX= can take one of these three values:
#   enforcing - SELinux security policy is enforced.
#   permissive - SELinux prints warnings instead of enforcing.
#   disabled - No SELinux policy is loaded.
SELINUX=disabled
# SELINUXTYPE= can take one of three values:
#   targeted - Targeted processes are protected,
#   minimum - Modification of targeted policy. Only selected processes are pr
#   mls - Multi Level Security protection.
SELINUXTYPE=targeted
```

更换阿里源，重建 yum 源缓存

```
[root@localhost wushuyue]# cd /etc/yum.repos.d/
[root@localhost yum.repos.d]# rm -rf *
[root@localhost yum.repos.d]# curl -o /etc/yum.repos.d/CentOS-Base.repo https://
mirrors.aliyun.com/repo/Centos-7.repo
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 2523 100 2523    0     0 10157      0  --:--:-- --:--:-- --:--:-- 10173
[root@localhost yum.repos.d]# yum makecache
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
* base: mirrors.aliyun.com
* extras: mirrors.aliyun.com
* updates: mirrors.aliyun.com
```

对系统进行升级

```
[root@localhost yum.repos.d]# yum -y update
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
```

安装OpenStack Stein的yum库

```
[root@localhost yum.repos.d]# yum -y install centos-release-openstack-stein
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
* base: mirrors.aliyun.com
* extras: mirrors.aliyun.com
* updates: mirrors.aliyun.com
```

修改四个文件配置，就是把 mirrorlist 注释掉，修改 baseurl 改为阿里源

```

GNU nano 2.3.1      File: CentOS-OpenStack-stein.repo      Modified
# CentOS-OpenStack-stein.repo
#
# Please see http://wiki.centos.org/SpecialInterestGroup/Cloud for more
# information

[centos-openstack-stein]
name=CentOS-7 - OpenStack stein
baseurl=http://mirrors.aliyun.com/\$contentdir/\$releasever/cloud/\$basearch/opens\$
#mirrorlist=http://mirrorlist.centos.org/?release=\$releasever&arch=\$basearch&re\$
gpgcheck=1
enabled=1

```

```

GNU nano 2.3.1      File: CentOS-Ceph-Nautilus.repo      Modified
# CentOS-Ceph-Nautilus.repo
#
# Please see https://wiki.centos.org/SpecialInterestGroup/Storage for more
# information

[centos-ceph-nautilus]
name=CentOS-$releasever - Ceph Nautilus
#mirrorlist=http://mirrorlist.centos.org/?release=\$releasever&arch=\$basearch&re\$
baseurl=http://mirrors.aliyun.com/\$contentdir/\$releasever/storage/\$basearch/cep\$
gpgcheck=1
enabled=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-CentOS-SIG-Storage

```

```

GNU nano 2.3.1      File: CentOS-QEMU-EV.repo      Modified
# CentOS-QEMU-EV.repo
#
# Please see http://wiki.centos.org/SpecialInterestGroup/Virtualization for more
# information

[centos-qemu-ev]
name=CentOS-$releasever - QEMU EV
#mirrorlist=http://mirrorlist.centos.org/?release=\$releasever&arch=\$basearch&re\$
baseurl=http://mirrors.aliyun.com/\$contentdir/\$releasever/virt/\$basearch/kvm-co\$
gpgcheck=1

```

```
GNU nano 2.3.1      File: CentOS-NFS-Ganesha-28.repo      Modified

# CentOS-NFS-Ganesha-28.repo
#
# Please see http://wiki.centos.org/SpecialInterestGroup/Storage for more
# information

[centos-nfs-ganesha28]
name=CentOS-$releasever - NFS Ganesha 2.8
#mirrorlist=http://mirrorlist.centos.org?arch=$basearch&release=$releasever&rep$
baseurl=https://mirrors.aliyun.com/$contentdir/$releasever/storage/$basearch/nf$
gpgcheck=1
enabled=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-CentOS-SIG-Storage
```

清理本地 yum 索引缓存 再重建索引缓存

```
[root@localhost yum.repos.d]# yum clean all && yum makecache
Loaded plugins: fastestmirror, langpacks
Cleaning repos: base centos-ceph-nautilus centos-nfs-ganesha28
                  : centos-openstack-stein centos-qemu-ev extras updates
Cleaning up list of fastest mirrors
```

安装 packstack 工具

```
[root@localhost ~]# yum -y install openstack-packstack
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
* base: mirrors.aliyun.com
```

部署 Openstack

```
complete.
[root@localhost ~]# packstack --allinone
Welcome to the Packstack setup utility

The installation log file is available at: /var/tmp/packstack/20260105-061550-hDCyT9/op
enstack-setup.log
Packstack changed given value  to required value /root/.ssh/id_rsa.pub
```

安装成功


```

Finalizing [ DONE ]

**** Installation completed successfully ****

Additional information:
* Parameter CONFIG_NEUTRON_L2_AGENT: You have chosen OVN neutron backend. Note that this backend does not support LBaaS, VPNaaS or FWaaS services. Geneve will be used as encapsulation method for tenant networks
* A new answerfile was created in: /root/packstack-answers-20260105-061550.txt
* Time synchronization installation was skipped. Please note that unsynchronized time on server instances might be problem for some OpenStack components.
* File /root/keystonerc_admin has been created on OpenStack client host 192.168.137.138. To use the command line tools you need to source the file.
* To access the OpenStack Dashboard browse to http://192.168.137.138/dashboard . Please, find your login credentials stored in the keystonerc_admin in your home directory.
* Because of the kernel update the host 192.168.137.138 requires reboot.
* The installation log file is available at: /var/tmp/packstack/20260105-061550-hDCyT9/openstack-setup.log
* The generated manifests are available at: /var/tmp/packstack/20260105-061550-hDCyT9/manifests

```

查看 OpenStack 管理员用户名和密码

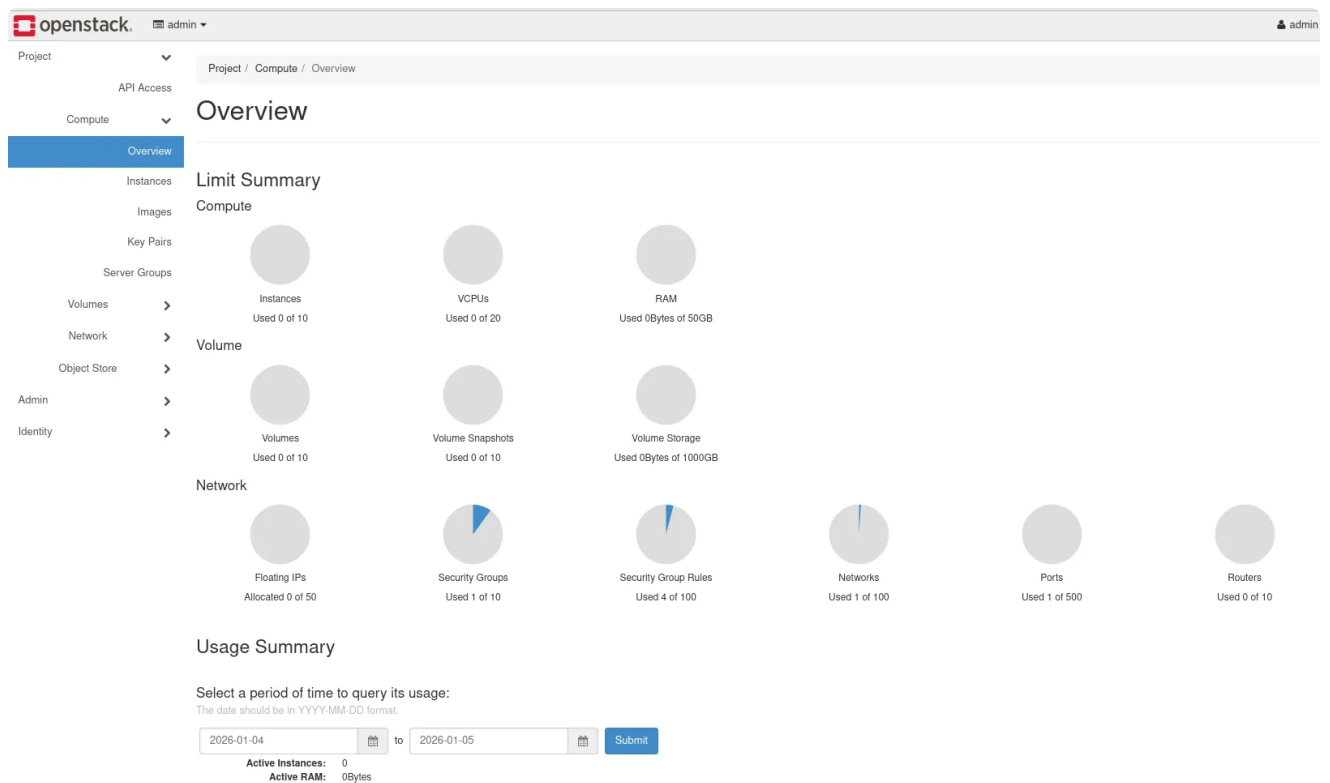
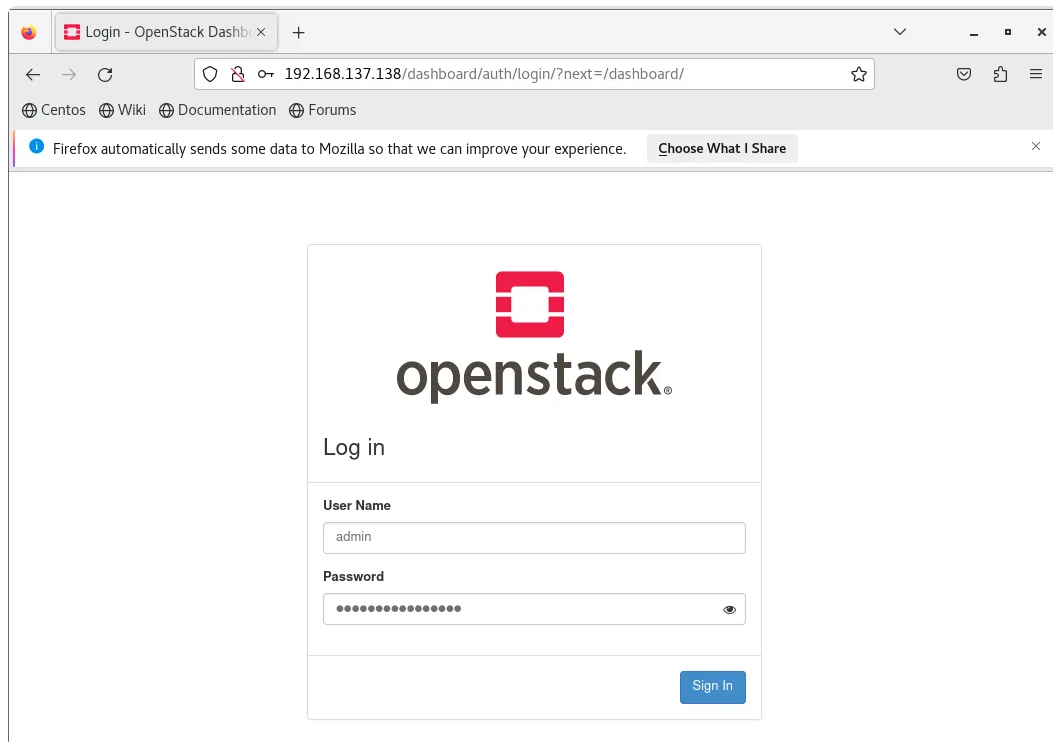
```

[root@localhost ~]# cat keystonerc_admin
unset OS_SERVICE_TOKEN
export OS_USERNAME=admin
export OS_PASSWORD='34f1f377ec3840ff'
export OS_REGION_NAME=RegionOne
export OS_AUTH_URL=http://192.168.137.138:5000/v3
export PS1='[\u@\h \W(keystone_admin)]\$ '

export OS_PROJECT_NAME=admin
export OS_USER_DOMAIN_NAME=Default
export OS_PROJECT_DOMAIN_NAME=Default
export OS_IDENTITY_API_VERSION=3

```

登录 Dashboard



获取管理员凭证，部署成功后，会在 root 用户目录下生成凭证文件
加载管理员凭证并验证凭证是否生效，如果命令返回所有服务列表，说明部署成功

```

[root@localhost ~]# cd
[root@localhost ~]# ls
anaconda-ks.cfg  keystone_admin  keystone_demo  original-ks.cfg  packstack-answers-20260105-061550.txt
[root@localhost ~]# source keystone_admin
[root@localhost ~(keystone_admin)]# openstack service list
+-----+-----+-----+
| ID | Name | Type |
+-----+-----+-----+
| 05791b9c14204d19b1957df844f39fd6 | keystone | identity |
| 0d7d5637d9194692b41d1fb14ebe48ed | nova | compute |
| 189a8e66142942f487a28845f3c6e125 | glance | image |
| 7111cb7f330948d79ad358ad678289fe | swift | object-store |
| 751b3b00bb754e5889b5acb35558531d | gnocchi | metric |
| 7f3f1c211d154020ba74d3e27ad48c18 | aodh | alarming |
| 830df958322848e385e6a860781370d8 | cinderv2 | volumev2 |
| 92423a41106d41639e23a20657416d79 | ceilometer | metering |
| 94a8ccd296c84bfbadd4a1e68d50791e | placement | placement |
| d6337b34688041eeaze32e0e12f2c980 | cinderv3 | volumev3 |
| f0866935c32b4a37889219710a97c911 | neutron | network |
+-----+-----+-----+

```

可以使用 openstack token issue 检测凭证是否生效

```

[root@localhost ~(keystone_admin)]# openstack token issue
+-----+-----+-----+
| Field | Value |
+-----+-----+-----+
| expires | 2026-01-05T15:55:25+0000 |
| id | gAAAAABpw9Dd8t3pSFYhTLPf0b-nvguby2Iv7r8leK_vPVsQRuExlmemge0k7CjjTwY5Q7YYZ12fSzDC9is03nSzF0r3RAnFG7m7MHFcIuIyezlaH9DjJJ0u_I8NB80waR0ov1M0i1TKoyJe01kwPzafp-80hurFd9m9nECggm0RbjbIGah3G0 |
| project_id | ebdee65ac8fb45379a2884d5d238ae7c |
| user_id | 3f40cbe575c545bd9beee485a4d994ae |
+-----+-----+-----+

```

网络架构设计与实现

查看现有默认资源，网络方面存在 private 和 public，路由有 router1

```

[root@localhost ~(keystone_admin)]# openstack network list
+-----+-----+-----+
| ID | Name | Subnets |
+-----+-----+-----+
| 8a52aec5-9c54-4a0b-93fa-862df56099f4 | private | 3cb7462f-8700-4fec-8334-be0ac83d84cf |
| f3ae204f-e066-453a-b58e-f8eed9974da5 | public | 59902ee4-9b58-4093-ac9d-b937dced2996 |
+-----+-----+-----+
[root@localhost ~(keystone_admin)]# openstack image list
+-----+-----+-----+
| ID | Name | Status |
+-----+-----+-----+
| 56487399-7f0f-4356-9a23-0368a81d7af7 | cirros | active |
+-----+-----+-----+
[root@localhost ~(keystone_admin)]# openstack flavor list
+-----+-----+-----+
| ID | Name | RAM | Disk | Ephemeral | VCPUs | Is Public |
+-----+-----+-----+
| 1 | m1.tiny | 512 | 1 | 0 | 1 | True |
| 2 | m1.small | 2048 | 20 | 0 | 1 | True |
| 3 | m1.medium | 4096 | 40 | 0 | 2 | True |
| 4 | m1.large | 8192 | 80 | 0 | 4 | True |
| 5 | m1.xlarge | 16384 | 160 | 0 | 8 | True |
+-----+-----+-----+
[root@localhost ~(keystone_admin)]# openstack router list
+-----+-----+-----+
| ID | Name | Status | State | Project |
+-----+-----+-----+
| a87c3965-7875-486c-b137-23b288719c58 | router1 | ACTIVE | UP | 76cd282f03b0478d9873d6b344ef645f |
+-----+-----+-----+

```

创建第二个内部网络

```

1  # 创建第二个内部网络
2  openstack network create internal-net2
3
4  # 创建第二个内部网络的子网
5  openstack subnet create \
6  --network internal-net2 \
7  --subnet-range 10.0.2.0/24 \
8  --gateway 10.0.2.1 \
9  --dns-nameserver 8.8.8.8 \
10 internal-subnet2
11
12 # 查看创建结果
13 openstack network list
14 openstack subnet list

```

```
[root@localhost ~(keystone_admin)]# openstack network create internal-net2
```

Field	Value
admin_state_up	UP
availability_zone_hints	
availability_zones	
created_at	2026-01-05T15:10:08Z
description	
dns_domain	None
id	bdac4caf-ae6b-47be-9ced-1d52a094a95c
ipv4_address_scope	None
ipv6_address_scope	None
is_default	False
is_vlan_transparent	None
location	Munch({'project': Munch({'domain_name': 'Default', 'domain_id': None, 'name': 'admin', 'id': 'u'ebdee65ac8fb45379a2884d5d238ae7c'}), 'cloud': '', 'region_name': 'RegionOne', 'zone': None})
mtu	1442
name	internal-net2
port_security_enabled	True
project_id	ebdee65ac8fb45379a2884d5d238ae7c
provider:network_type	geneve
provider:physical_network	None
provider:segmentation_id	11
qos_policy_id	None
revision_number	2
router:external	Internal
segments	None
shared	False
status	ACTIVE
subnets	
tags	
updated_at	2026-01-05T15:10:08Z

```

[root@localhost ~(keystone_admin)]# openstack subnet create \
> --network internal-net2 \
> --subnet-range 10.0.2.0/24 \
> --gateway 10.0.2.1 \
> --dns-nameserver 8.8.8.8 \
> internal-subnet2

```

Field	Value
allocation_pools	10.0.2.2-10.0.2.254
cidr	10.0.2.0/24
created_at	2026-01-05T15:10:28Z
description	
dns_nameservers	8.8.8.8
enable_dhcp	True
gateway_ip	10.0.2.1
host_routes	
id	0581e432-675e-4a49-948e-b94bbcfaf10f1
ip_version	4
ipv6_address_mode	None
ipv6_ra_mode	None
location	Munch({'project': Munch({'domain_name': 'Default', 'domain_id': None, 'name': 'admin', 'id': 'u'ebdee65ac8fb45379a2884d5d238ae7c'}), 'cloud': '', 'region_name': 'RegionOne', 'zone': None})
name	internal-subnet2
network_id	bdac4caf-ae6b-47be-9ced-1d52a094a95c
prefix_length	None
project_id	ebdee65ac8fb45379a2884d5d238ae7c
revision_number	0
segment_id	None
service_types	
subnetpool_id	None
tags	
updated_at	2026-01-05T15:10:28Z

查看创建结果


```
[root@localhost ~(keystone_admin)]# openstack network list
+-----+-----+-----+
| ID | Name | Subnets |
+-----+-----+-----+
| 8a52aec5-9c54-4a0b-93fa-862df56099f4 | private | 3cb7462f-8700-4fec-8334-be0ac83d84cf |
| bdac4caf-ae6b-47be-9ced-1d52a094a95c | internal-net2 | 0581e432-675e-4a49-948e-894bbcf10f1 |
| f3ae204f-e066-453a-b58e-f8eed9974da5 | public | 59902ee4-9b58-4093-ac9d-b937dced2996 |
+-----+-----+-----+
[root@localhost ~(keystone_admin)]# openstack subnet list
+-----+-----+-----+-----+
| ID | Name | Network | Subnet |
+-----+-----+-----+-----+
| 0581e432-675e-4a49-948e-894bbcf10f1 | internal-subnet2 | bdac4caf-ae6b-47be-9ced-1d52a094a95c | 10.0.2.0/24 |
| 3cb7462f-8700-4fec-8334-be0ac83d84cf | private_subnet | 8a52aec5-9c54-4a0b-93fa-862df56099f4 | 10.0.0.0/24 |
| 59902ee4-9b58-4093-ac9d-b937dced2996 | public_subnet | f3ae204f-e066-453a-b58e-f8eed9974da5 | 172.24.4.0/24 |
+-----+-----+-----+-----+
[root@localhost ~(keystone_admin)]#
```

创建路由器并连接网络

Python

```
1 # 1. 创建路由器
2 openstack router create my-router
3
4 # 2. 设置外部网关（使用现有的public网络）
5 openstack router set my-router --external-gateway public
6
7 # 3. 连接两个内部子网到路由器
8 # 先查看private网络的子网ID
9 PRIVATE_SUBNET_ID=$(openstack subnet list --network private -c ID -f value
10 )
11 # 连接两个子网
12 openstack router add subnet my-router $PRIVATE_SUBNET_ID
13 openstack router add subnet my-router internal-subnet2
14
15 # 4. 查看路由器状态
16 openstack router show my-router
17
```

```
[root@localhost ~(keystone_admin)]# openstack router create my-router
+-----+-----+
| Field | Value |
+-----+-----+
| admin_state_up | UP |
| availability_zone_hints | None |
| availability_zones | None |
| created_at | 2026-01-05T15:12:18Z |
| description | |
| external_gateway_info | None |
| flavor_id | None |
| id | 8f533c5e-4042-40d4-bfaf-8e22b845b75e |
| location | Munch({'project': Munch({'domain_name': 'Default', 'domain_id': None, 'name': 'admin', 'id': 'u'ebdee65ac8fb45379a2884d5d238ae7c'}), 'cloud': '', 'region_name': 'RegionOne', 'zone': None}) |
| name | my-router |
| project_id | ebdee65ac8fb45379a2884d5d238ae7c |
| revision_number | 0 |
| routes | |
| status | ACTIVE |
| tags | |
| updated_at | 2026-01-05T15:12:18Z |
+-----+-----+

[root@localhost ~(keystone_admin)]# openstack router set my-router --external-gateway public
[root@localhost ~(keystone_admin)]# PRIVATE_SUBNET_ID=$(openstack subnet list --network private -c ID -f value)
[root@localhost ~(keystone_admin)]# openstack router add subnet my-router $PRIVATE_SUBNET_ID
[root@localhost ~(keystone_admin)]# openstack router add subnet my-router internal-subnet2
[root@localhost ~(keystone_admin)]# openstack router show my-router
+-----+-----+
| Field | Value |
+-----+-----+
| admin_state_up | UP |
| availability_zone_hints | None |
| availability_zones | None |
| created_at | 2026-01-05T15:12:18Z |
| description | |
| external_gateway_info | {'network_id': 'f3ae204f-e066-453a-b58e-f8eed9974da5', 'enable_snat': true, 'external_fixed_ips': [{'subnet_id': '59902ee4-9b58-4093-ac9d-b937dced2996', 'ip_address': '172.24.4.160'}]} |
| id | None |
| interfaces_info | [{'subnet_id': '0581e432-675e-4a49-948e-894bbcf10f1', 'ip_address': '10.0.2.1', 'port_id': 'caaaa9fd-4f52-42bb-86d5-fd0ef85e0586'}] |
| location | Munch({'project': Munch({'domain_name': 'Default', 'domain_id': None, 'name': 'admin', 'id': 'u'ebdee65ac8fb45379a2884d5d238ae7c'}), 'cloud': '', 'region_name': 'RegionOne', 'zone': None}) |
| name | my-router |
| project_id | ebdee65ac8fb45379a2884d5d238ae7c |
| revision_number | 3 |
| routes | |
| status | ACTIVE |
| tags | |
| updated_at | 2026-01-05T15:13:04Z |
+-----+-----+
```

创建安全组

Python |

```
1 # 创建安全组（允许ICMP和SSH）
2 openstack security group create --description "Basic security group" basic
  -sg
3
4 # 添加ICMP规则（允许ping）
5 openstack security group rule create \
6     --protocol icmp \
7     --remote-ip 0.0.0.0/0 \
8     basic-sg
9
10 # 添加SSH规则
11 openstack security group rule create \
12     --protocol tcp \
13     --dst-port 22 \
14     --remote-ip 0.0.0.0/0 \
15     basic-sg
16
17 # 查看安全组规则
18 openstack security group rule list basic-sg
```

```
[root@localhost ~(keystone_admin)]# openstack security group create --description "Basic security group" basic-sg
+-----+-----+
| Field | Value |
+-----+-----+
| created_at | 2026-01-05T15:15:14Z |
| description | Basic security group |
| id | e5e695eb-eb04-4bcf-a530-0cdf0194dce0 |
| location | Munch({'project': Munch({'domain_name': 'Default', 'domain_id': None, 'name': 'admin', 'id': 'u'ebdee65ac8fb45379a2884d5d238ae7c'}), 'cloud': '', 'region_name': 'RegionOne', 'zone': None}) |
| name | basic-sg |
| project_id | ebdee65ac8fb45379a2884d5d238ae7c |
| revision_number | 1 |
| rules | created_at='2026-01-05T15:15:14Z', direction='egress', ethertype='IPv6', id='7f44e302-a908-4d3b-aa30-d4570466ea51', updated_at='2026-01-05T15:15:14Z'
| created_at='2026-01-05T15:15:14Z', direction='egress', ethertype='IPv4', id='da089e4d-50f0-463b-a78c-672e7105df1b', updated_at='2026-01-05T15:15:14Z' |
| tags | [] |
| updated_at | 2026-01-05T15:15:14Z |
+-----+-----+
```

```
[root@localhost ~(keystone_admin)]# openstack security group rule create \
> --protocol icmp \
> --remote-ip 0.0.0.0/0 \
> basic-sg
+-----+
| Field | Value |
+-----+
| created_at | 2026-01-05T15:15:30Z |
| description | |
| direction | ingress |
| ether_type | IPv4 |
| id | 07d07838-2234-4236-9842-c21ef70d6494 |
| location | Munch({'project': Munch({'domain_name': 'Default', 'domain_id': None, 'name': 'admin', 'id': 'u'ebdee65ac8fb45379a2884d5d238ae7c'}), 'cloud': '', 'region_name': 'RegionOne', 'zone': None}) |
| name | None |
| port_range_max | None |
| port_range_min | None |
| project_id | ebdee65ac8fb45379a2884d5d238ae7c |
| protocol | icmp |
| remote_group_id | None |
| remote_ip_prefix | 0.0.0.0/0 |
| revision_number | 0 |
| security_group_id | e5e695eb-e804-4bcf-a530-0cdf0194dce0 |
| tags | [] |
| updated_at | 2026-01-05T15:15:30Z |
+-----+

[root@localhost ~(keystone_admin)]# openstack security group rule create \
> --protocol tcp \
> --dst-port 22 \
> --remote-ip 0.0.0.0/0 \
> basic-sg
+-----+
| Field | Value |
+-----+
| created_at | 2026-01-05T15:15:36Z |
| description | |
| direction | ingress |
| ether_type | IPv4 |
| id | 93293eb6-5067-415a-a5d0-321d2bb485d5 |
| location | Munch({'project': Munch({'domain_name': 'Default', 'domain_id': None, 'name': 'admin', 'id': 'u'ebdee65ac8fb45379a2884d5d238ae7c'}), 'cloud': '', 'region_name': 'RegionOne', 'zone': None}) |
| name | None |
| port_range_max | 22 |
| port_range_min | 22 |
| project_id | ebdee65ac8fb45379a2884d5d238ae7c |
| protocol | tcp |
| remote_group_id | None |
| remote_ip_prefix | 0.0.0.0/0 |
| revision_number | 0 |
| security_group_id | e5e695eb-e804-4bcf-a530-0cdf0194dce0 |
| tags | [] |
| updated_at | 2026-01-05T15:15:36Z |
+-----+

+-----+
| ID | IP Protocol | IP Range | Port Range | Remote Security Group |
+-----+
| 07d07838-2234-4236-9842-c21ef70d6494 | icmp | 0.0.0.0/0 | | None |
| 7f44e302-a908-4d3b-aa30-d4570466ea51 | None | None | | None |
| 93293eb6-5067-415a-a5d0-321d2bb485d5 | tcp | 0.0.0.0/0 | 22:22 | None |
| da089e4d-50f0-463b-a78c-672e7105df1b | None | None | | None |
+-----+
```

创建密钥对

```
1 # 创建密钥对（如果已存在可跳过）
2 openstack keypair create mykey > mykey.pem
3 chmod 600 mykey.pem
4
5 # 查看密钥对
6 openstack keypair list
```

```
[root@localhost ~(keystone_admin)]# openstack keypair create mykey > mykey.pem
[root@localhost ~(keystone_admin)]# chmod 600 mykey.pem
[root@localhost ~(keystone_admin)]# openstack keypair list
+-----+
| Name | Fingerprint |
+-----+
| mykey | 7d:92:5c:6f:46:a9:c4:c0:f7:4a:5e:01:fe:70:df:0c |
+-----+
[root@localhost ~(keystone_admin)]#
```

创建两台云主机

创建第一台云主机（在 private 网络）

```

1  # 创建第一台云主机 (使用private网络)
2  openstack server create \
3      --image cirros \
4      --flavor m1.tiny \
5      --network private \
6      --security-group basic-sg \
7      --key-name mykey \
8      vm1
9
10 openstack server create \
11     --image cirros-0.5.2 \
12     --flavor m1.tiny \
13     --network private \
14     --security-group basic-sg \
15     --key-name mykey \
16     vm1
17
18
19
20 # 分配浮动IP给vm1
21 FLOATING_IP1=$(openstack floating ip create public -c floating_ip_address
    -f value)
22 openstack server add floating ip vm1 $FLOATING_IP1

```

```

[root@localhost ~(keystone_admin)]# openstack server create \
> --image cirros \
> --flavor m1.tiny \
> --network private \
> --security-group basic-sg \
> --key-name mykey \
> vm1
+-----+
| Field                                | Value                                |
+-----+-----+
| OS-DCF:diskConfig                    | MANUAL                              |
| OS-EXT-AZ:availability_zone          |                                     |
| OS-EXT-SRV-ATTR:host                 | None                                |
| OS-EXT-SRV-ATTR:hypervisor_hostname | None                                |
| OS-EXT-SRV-ATTR:instance_name        |                                     |
| OS-EXT-STS:power_state                | NOSTATE                             |
| OS-EXT-STS:task_state                 | scheduling                           |
| OS-EXT-STS:vm_state                   | building                             |
| OS-SRV-USG:launched_at                | None                                |
| OS-SRV-USG:terminated_at              | None                                |
| accessIPv4                            |                                     |
| accessIPv6                            |                                     |
| addresses                             |                                     |
| adminPass                             | WppJYgFwd8S8                        |
| config_drive                          |                                     |
| created                               | 2026-01-05T15:18:26Z                |
| flavor                                | m1.tiny (1)                         |
| hostId                                |                                     |
| id                                    | 4c63b50b-5de1-475f-b682-b26795d07c43 |
| image                                 | cirros (56487399-7f0f-4356-9a23-0368a81d7af7) |
| key_name                              | mykey                               |
| name                                  | vm1                                  |
| progress                              | 0                                    |
| project_id                            | ebdee65ac8fb45379a2884d5d238ae7c    |
| properties                            |                                     |
| security_groups                       | name='e5e695eb-eb04-4bcf-a530-0cdf0194dce0' |
| status                                | BUILD                               |
| updated                               | 2026-01-05T15:18:26Z                |
| user_id                               | 3f40cbe575c545bd9beee485a4d994ae    |
| volumes_attached                      |                                     |
+-----+-----+
[root@localhost ~(keystone_admin)]# FLOATING_IP1=$(openstack floating ip create public -c floating_ip_address -f value)
[root@localhost ~(keystone_admin)]# openstack server add floating ip vm1 $FLOATING_IP1

```

创建第二台云主机（在 internal-net2 网络）

Python |

```
1  # 创建第二台云主机（使用新创建的internal-net2网络）
2  openstack server create \
3      --image cirros \
4      --flavor m1.tiny \
5      --network internal-net2 \
6      --security-group basic-sg \
7      --key-name mykey \
8      vm2
9
10 # 分配浮动IP给vm2
11 FLOATING_IP2=$(openstack floating ip create public -c floating_ip_address
    -f value)
12 openstack server add floating ip vm2 $FLOATING_IP2
```

```
[root@localhost ~(keystone_admin)]# openstack server add floating ip vm2 $FLOATING_IP2
[root@localhost ~(keystone_admin)]# openstack server create \
> --image cirros \
> --flavor m1.tiny \
> --network internal-net2 \
> --security-group basic-sg \
> --key-name mykey \
> vm2
+-----+-----+
| Field | Value |
+-----+-----+
| OS-DCF:diskConfig | MANUAL |
| OS-EXT-AZ:availability_zone | None |
| OS-EXT-SRV-ATTR:host | None |
| OS-EXT-SRV-ATTR:hypervisor_hostname | None |
| OS-EXT-SRV-ATTR:instance_name | None |
| OS-EXT-STS:power_state | NOSTATE |
| OS-EXT-STS:task_state | scheduling |
| OS-EXT-STS:vm_state | building |
| OS-SRV-USG:launched_at | None |
| OS-SRV-USG:terminated_at | None |
| accessIPv4 | |
| accessIPv6 | |
| addresses | |
| adminPass | y9wTyxjGrQUs |
| config_drive | |
| created | 2026-01-05T15:19:21Z |
| flavor | m1.tiny (1) |
| hostId | |
| id | ea55989d-72dd-4dd0-8eb3-ba7ae05f8ad1 |
| image | cirros (56487399-7f0f-4356-9a23-0368a81d7af7) |
| key_name | mykey |
| name | vm2 |
| progress | 0 |
| project_id | ebdee65ac8fb45379a2884d5d238ae7c |
| properties | |
| security_groups | name='e5e695eb-eb04-4bcf-a530-0cdf0194dce0' |
| status | BUILD |
| updated | 2026-01-05T15:19:21Z |
| user_id | 3f40cbe575c545bd9beee485a4d994ae |
| volumes_attached | |
+-----+-----+
[root@localhost ~(keystone_admin)]# FLOATING_IP2=$(openstack floating ip create public -c floating_ip_address -f value)
[root@localhost ~(keystone_admin)]# openstack server add floating ip vm2 $FLOATING_IP2
[root@localhost ~(keystone_admin)]#
```

等待并检查虚拟机状态

```

1  # 等待虚拟机启动 (约30-60秒)
2  echo "等待虚拟机启动..."
3  sleep 30
4
5  # 检查虚拟机状态
6  openstack server list
7
8  # 查看详细信息
9  echo "VM1详细信息: "
10 openstack server show vm1 -c name -c status -c addresses
11
12 echo "VM2详细信息: "
13 openstack server show vm2 -c name -c status -c addresses

```

```

[root@localhost ~(keystone_admin)]# openstack server add floating ip vm2 $FLOATING_IP2
[root@localhost ~(keystone_admin)]# openstack server list
+-----+-----+-----+-----+-----+-----+
| ID | Name | Status | Networks | Image | Flavor |
+-----+-----+-----+-----+-----+-----+
| ea55989d-72dd-4dd0-8eb3-ba7ae05f8ad1 | vm2 | ACTIVE | internal-net2=10.0.2.87, 172.24.4.213 | cirros | m1.tiny |
| 4c63b50b-5de1-475f-b682-b26795d07c43 | vm1 | ACTIVE | private=10.0.0.96, 172.24.4.26 | cirros | m1.tiny |
+-----+-----+-----+-----+-----+-----+
[root@localhost ~(keystone_admin)]# openstack server show vm1 -c name -c status -c addresses
+-----+-----+
| Field | Value |
+-----+-----+
| addresses | private=10.0.0.96, 172.24.4.26 |
| name | vm1 |
| status | ACTIVE |
+-----+-----+
[root@localhost ~(keystone_admin)]# openstack server show vm2 -c name -c status -c addresses
+-----+-----+
| Field | Value |
+-----+-----+
| addresses | internal-net2=10.0.2.87, 172.24.4.213 |
| name | vm2 |
| status | ACTIVE |
+-----+-----+
[root@localhost ~(keystone_admin)]#

```

网络连通性测试

```

1  # 获取虚拟机内部IP地址
2  VM1_INTERNAL_IP=$(openstack server show vm1 -c addresses -f value | grep -o
E '10\.0\.[0-9]+\.[0-9]+' | head -1)
3  VM2_INTERNAL_IP=$(openstack server show vm2 -c addresses -f value | grep -o
E '10\.0\.[0-9]+\.[0-9]+' | head -1)
4
5  echo "VM1内部IP: $VM1_INTERNAL_IP"
6  echo "VM2内部IP: $VM2_INTERNAL_IP"
7  echo "VM1浮动IP: $FLOATING_IP1"
8  echo "VM2浮动IP: $FLOATING_IP2"

```

```

[root@localhost ~](keystone_admin)]# VM1_INTERNAL_IP=$(openstack server show vm1 -c addresses -f value | grep -oE '10\.0\.[0-9]+\.[0-9]+' | head -1)
[root@localhost ~](keystone_admin)]# VM2_INTERNAL_IP=$(openstack server show vm2 -c addresses -f value | grep -oE '10\.0\.[0-9]+\.[0-9]+' | head -1)
[root@localhost ~](keystone_admin)]# echo "VM1内部IP: $VM1_INTERNAL_IP"
VM1内部IP: 10.0.0.96
[root@localhost ~](keystone_admin)]# echo "VM2内部IP: $VM2_INTERNAL_IP"
VM2内部IP: 10.0.2.87
[root@localhost ~](keystone_admin)]# echo "VM1浮动IP: $FLOATING_IP1"
VM1浮动IP: 172.24.4.26
[root@localhost ~](keystone_admin)]# echo "VM2浮动IP: $FLOATING_IP2"
VM2浮动IP: 172.24.4.213

```

使用控制台测试

Python

```

1  # 1. 创建测试脚本
2  cat > test_network.sh << 'EOF'
3  #!/bin/bash
4  echo "=== 网络联通性测试 ==="
5  echo "1. 测试VM1网络连通性"
6  echo "2. 测试VM2网络连通性"
7  echo "3. 测试VM1和VM2之间的连通性"
8  echo ""
9  echo "由于cirros不支持SSH, 请通过以下方式测试: "
10 echo "1. 在Dashboard中打开VM1的控制台"
11 echo "2. 登录 (用户名: cirros, 密码: gocubsgo) "
12 echo "3. 执行: ping $VM2_INTERNAL_IP"
13 echo "4. 执行: ping 8.8.8.8"
14 echo ""
15 echo "同理测试VM2"
16 EOF
17
18 chmod +x test_network.sh
19 ./test_network.sh

```

```

[root@localhost ~]# ping 172.24.4.18
PING 172.24.4.18 (172.24.4.18) 56(84) bytes of data.
64 bytes from 172.24.4.18: icmp_seq=1 ttl=63 time=1.51 ms
64 bytes from 172.24.4.18: icmp_seq=2 ttl=63 time=0.364 ms
64 bytes from 172.24.4.18: icmp_seq=3 ttl=63 time=0.322 ms

```

云主机生命周期管理

1 查看云主机列表

Python

```

1  # 查看云主机列表 (带详细字段)
2  openstack server list -c Name -c Status -c PowerState -c TaskState -c Host
   -c Networks

```



```

+-----+-----+-----+-----+-----+-----+
[root@localhost ~(keystone_admin)]# openstack server list -c Name -c Status -c PowerState -c TaskState -c Host -c Networks
+-----+-----+-----+-----+-----+-----+
| Name | Status | Networks |
+-----+-----+-----+-----+-----+
| vm2 | ACTIVE | internal-net2=10.0.2.82, 172.24.4.151 |
| vm1 | ACTIVE | private=10.0.0.227, 172.24.4.222 |
+-----+-----+-----+-----+-----+

```

2 启停操作

Python |

```

1  openstack server stop vm1
2
3  # 检查状态
4  openstack server show vm1 -c status -c power_state
5
6
7  # 启动云主机
8  openstack server start vm1
9
10 # 检查状态
11 openstack server show vm1 -c status -c power_state

```

```

+-----+
[root@localhost ~(keystone_admin)]# openstack server stop vm1
Cannot 'stop' instance 6aee589e-1541-4fd7-934b-059a471cc2a1 while it is in task_state powering-off (HTTP 409) (Request-ID: req-502c9d37-03a1-4460-b37c-bb0514f4cc42)
[root@localhost ~(keystone_admin)]# openstack server show vm1 -c status -c power_state
+-----+-----+
| Field | Value |
+-----+-----+
| status | SHUTOFF |
+-----+-----+
-

+-----+
[root@localhost ~(keystone_admin)]# openstack server start vm1
[root@localhost ~(keystone_admin)]# openstack server show vm1 -c status -c power_state
+-----+-----+
| Field | Value |
+-----+-----+
| status | ACTIVE |
+-----+-----+
[root@localhost ~(keystone_admin)]#

```

3 重启操作

Python |

```

1  # 软重启 (graceful shutdown)
2  openstack server reboot --soft vm1
3
4  # 等待重启
5  openstack server show vm1 -c status
6
7  # 硬重启
8  openstack server reboot --hard vm1
9
10 # 等待重启
11 openstack server show vm1 -c status

```

```

+-----+
[root@localhost ~(keystone_admin)]# openstack server reboot --soft vm1
[root@localhost ~(keystone_admin)]# openstack server show vm1 -c status
+-----+
| Field | Value |
+-----+
| status | REBOOT |
+-----+
[root@localhost ~(keystone_admin)]#
+-----+
[root@localhost ~(keystone_admin)]# openstack server reboot --hard vm1
[root@localhost ~(keystone_admin)]# openstack server show vm1 -c status
+-----+
| Field | Value |
+-----+
| status | ACTIVE |
+-----+
[root@localhost ~(keystone_admin)]# openstack server show vm1 -c status

```

4 暂停/恢复操作

Python |

```

1  # 暂停云主机（内存状态保存到磁盘）
2  openstack server pause vm1
3
4  # 检查状态
5  openstack server show vm1 -c status -c power_state
6
7  # 恢复云主机
8  openstack server unpause vm1
9
10 # 检查状态
11 openstack server show vm1 -c status -c power_state

```

```

+-----+
[root@localhost ~(keystone_admin)]# openstack server pause vm1
[root@localhost ~(keystone_admin)]# openstack server show vm1 -c status -c power_state
+-----+
| Field | Value |
+-----+
| status | PAUSED |
+-----+
[root@localhost ~(keystone_admin)]#
+-----+
[root@localhost ~(keystone_admin)]# openstack server unpause vm1
[root@localhost ~(keystone_admin)]# openstack server show vm1 -c status -c power_state
+-----+
| Field | Value |
+-----+
| status | ACTIVE |
+-----+
[root@localhost ~(keystone_admin)]#

```

5 挂起/恢复操作

```

1  # 挂起云主机（类似休眠，内存状态写入磁盘）
2  openstack server suspend vm1
3
4  # 检查状态
5  openstack server show vm1 -c status -c power_state
6
7  # 恢复云主机
8  openstack server resume vm1
9
10 # 检查状态
11 openstack server show vm1 -c status -c power_state

```

```

[root@localhost ~(keystone_admin)]# openstack server suspend vm1
[root@localhost ~(keystone_admin)]# openstack server show vm1 -c status -c power_state
+-----+-----+
| Field | Value |
+-----+-----+
| status | SUSPENDED |
+-----+-----+
[root@localhost ~(keystone_admin)]#
[root@localhost ~(keystone_admin)]# openstack server resume vm1
[root@localhost ~(keystone_admin)]# openstack server show vm1 -c status -c power_state
+-----+-----+
| Field | Value |
+-----+-----+
| status | ACTIVE |
+-----+-----+
[root@localhost ~(keystone_admin)]#

```

6 锁定/解锁操作

```

1  # 锁定云主机（防止意外操作）
2  openstack server lock vm1
3
4  # 尝试删除（应该失败）
5  openstack server delete vm1 2>/dev/null && echo "删除成功" || echo "删除失败
   （符合预期，云主机被锁定）"
6
7  # 检查锁定状态
8  openstack server show vm1 -c locked
9
10 # 解锁云主机
11 openstack server unlock vm1
12
13 # 再次检查锁定状态
14 openstack server show vm1 -c locked

```

```

root@localhost ~(keystone_admin)]# openstack server lock vm1
root@localhost ~(keystone_admin)]#

```

```
[root@localhost ~(keystone_admin)]# openstack server add floating ip vm1 $FLOATING_IP1
[root@localhost ~(keystone_admin)]# openstack server lock vm1
[root@localhost ~(keystone_admin)]# openstack server show vm1 -c locked
No recognized column names in ['locked']. Recognized columns are ('OS-DCF:diskConfig', 'OS-EXT-AZ:availability_zone', 'OS-EXT-SRV-ATTR:host', 'OS-EXT-SRV-ATTR:hypervisor_hostname', 'OS-EXT-SRV-ATTR:instance_name', 'OS-EXT-STS:power_state', 'OS-EXT-STS:task_state', 'OS-EXT-STS:vm_state', 'OS-SRV-UG:launched_at', 'OS-SRV-UG:terminated_at', 'accessIPv4', 'accessIPv6', 'addresses', 'config_drive', 'created', 'flavor', 'hostId', 'id', 'image', 'key_name', 'name', 'progress', 'project_id', 'properties', 'security_groups', 'status', 'updated', 'user_id', 'volumes_attached').
[root@localhost ~(keystone_admin)]#
```

	Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
☐	vm1	cirros	10.0.0.22, 172.24.4.222	m1.tiny	mykey	Active	 nova	None	Running	0 minutes	Create Snapshot

```
File Edit View Search Terminal Help
[root@localhost ~(keystone_admin)]# openstack server unlock vm1
[root@localhost ~(keystone_admin)]# openstack server show vm1 -c locked
No recognized column names in ['locked']. Recognized columns are ('OS-DCF:diskConfig', 'OS-EXT-AZ:availability_zone', 'OS-EXT-SRV-ATTR:host', 'OS-EXT-SRV-ATTR:hypervisor_hostname', 'OS-EXT-SRV-ATTR:instance_name', 'OS-EXT-STS:power_state', 'OS-EXT-STS:task_state', 'OS-EXT-STS:vm_state', 'OS-SRV-UG:launched_at', 'OS-SRV-UG:terminated_at', 'accessIPv4', 'accessIPv6', 'addresses', 'config_drive', 'created', 'flavor', 'hostId', 'id', 'image', 'key_name', 'name', 'progress', 'project_id', 'properties', 'security_groups', 'status', 'updated', 'user_id', 'volumes_attached').
[root@localhost ~(keystone_admin)]#
```

7 调整规格

Python

```
1 # 查看当前规格
2 openstack server show vm1 -c flavor
3
4 # 停止云主机 (resize需要云主机停止或特殊状态)
5 openstack server stop vm1
6
7 # 调整到m1.small规格
8 openstack server resize --flavor m1.small vm1
9
10 # 等待resize完成
11 # 确认resize
12 openstack server resize confirm vm1
13
14 # 启动云主机
15 openstack server start vm1
16
17 # 验证新规格
18 openstack server show vm1 -c flavor -c status
```

```
a , properties , security_groups , status , updated , user_id , volumes_attached ,
[root@localhost ~(keystone_admin)]# openstack server show vm1 -c flavor
+-----+-----+
| Field | Value |
+-----+-----+
| flavor | m1.tiny (1) |
+-----+-----+
[root@localhost ~(keystone_admin)]# openstack server stop vm1
[root@localhost ~(keystone_admin)]# openstack server show vm1 -c status -c power_state
+-----+-----+
| Field | Value |
+-----+-----+
| status | SHUTOFF |
+-----+-----+
[root@localhost ~(keystone_admin)]# openstack server resize --flavor m1.small vm1
"Resize in progress. You can cancel the operation by running 'openstack server resize cancel vm1'."
```

Instance ID = Filter  Launch Instance  Locked Instances Mo

Displaying 2 items

	Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
☐	vm1	cirros	10.0.0.22, 172.24.4.222	m1.small	mykey	Confirm or Revert Resize/Migrate	us-east-1 nova	None	Shut Down	5 minutes	Confirm Resize?

从快照重建新云主机

Python |

```
1 # 从快照创建新云主机
2 openstack server create \
3     --image vm2-snapshot \
4     --flavor m1.tiny \
5     --network private \
6     --security-group basic-sg \
7     --key-name mykey \
8     vm2-clone
9
10 # 分配浮动IP
11 FLOAT_CLONE=$(openstack floating ip create public -c floating_ip_address -
12 f value)
13
14 # 等待启动
15 openstack server show vm2-clone -c name -c status -c addresses
```

```
+-----+
[root@localhost ~(keystone_admin)]# openstack server start vm2
[root@localhost ~(keystone_admin)]# openstack server create \
> --image vm2-snapshot \
> --flavor m1.tiny \
> --network private \
> --security-group basic-sg \
> --key-name mykey \
> vm2-clone
+-----+
| Field | Value |
+-----+
| OS-DCF:diskConfig | MANUAL |
| OS-EXT-AZ:availability_zone | None |
| OS-EXT-SRV-ATTR:host | None |
| OS-EXT-SRV-ATTR:hypervisor_hostname | None |
| OS-EXT-SRV-ATTR:instance_name | None |
| OS-EXT-STS:power_state | NOSTATE |
| OS-EXT-STS:task_state | scheduling |
| OS-EXT-STS:vm_state | building |
| OS-SRV-USG:launched_at | None |
| OS-SRV-USG:terminated_at | None |
| accessIPv4 | |
| accessIPv6 | |
| addresses | |
| adminPass | h2Rx6pb4mXgv |
| config_drive | |
| created | 2026-01-06T12:06:00Z |
| flavor | m1.tiny (1) |
| hostId | |
| id | 8ba08ac2-b5ec-44f4-bcdf-99637164d1bd |
| image | vm2-snapshot (631be80b-883b-48d1-9b69-6992b116f287) |
| key_name | mykey |
| name | vm2-clone |
| progress | 0 |
| project_id | bb7a5f3a824742988fdb998d0c1a68fb |
| properties | |
| security_groups | name='ca676a1f-90b0-40f6-9196-61f6bc419899' |
| status | BUILD |
| updated | 2026-01-06T12:06:00Z |
```

```

+-----+
[root@localhost ~(keystone_admin)]# FLOAT_CLONE=$(openstack floating ip create public -c floating_ip_address -f value)
[root@localhost ~(keystone_admin)]# openstack server add floating ip vm2-clone $FLOAT_CLONE
[root@localhost ~(keystone_admin)]# openstack server show vm2-clone -c name -c status -c addresses
+-----+
| Field | Value |
+-----+
| addresses | private=10.0.0.224, 172.24.4.210 |
| name | vm2-clone |
| status | ACTIVE |
+-----+

```

使用快照恢复现有云主机

```

1  # 方法：删除原云主机，用快照重建
2  ORIGINAL_VM="vm2"
3  SNAPSHOT_NAME="vm2-snapshot"
4
5  # 获取原云主机信息
6  ORIGINAL_NETWORK=$(openstack server show $ORIGINAL_VM -c addresses -f value | cut -d= -f1)
7  ORIGINAL_FLOAT_IP=$(openstack floating ip list --port $(openstack port list --server $ORIGINAL_VM -c ID -f value) -c "Floating IP Address" -f value)
8
9  echo "原云主机信息："
10 echo "- 名称：$ORIGINAL_VM"
11 echo "- 网络：$ORIGINAL_NETWORK"
12 echo "- 浮动IP：$ORIGINAL_FLOAT_IP"
13
14 # 删除原云主机
15 openstack server delete $ORIGINAL_VM
16
17 # 从快照重建云主机
18 openstack server create \
19     --image $SNAPSHOT_NAME \
20     --flavor m1.tiny \
21     --network $ORIGINAL_NETWORK \
22     --security-group basic-sg \
23     --key-name mykey \
24     $ORIGINAL_VM
25
26 # 重新分配原浮动IP
27 openstack server add floating ip $ORIGINAL_VM $ORIGINAL_FLOAT_IP
28
29 # 等待启动
30 openstack server show $ORIGINAL_VM -c name -c status -c addresses

```



```

+-----+
[root@localhost ~(keystone_admin)]# ORIGINAL_VM="vm2"
[root@localhost ~(keystone_admin)]# SNAPSHOT_NAME="vm2-snapshot"
[root@localhost ~(keystone_admin)]# openstack server show $ORIGINAL_VM -c addresses -f value | cut -d= -f1)
[root@localhost ~(keystone_admin)]# ORIGINAL_FLOAT_IP=$(openstack floating ip list --port $(openstack port list --server $ORIGINAL_VM -c ID
ess" -f value)
[root@localhost ~(keystone_admin)]# echo "原云主机信息："
原云主机信息：
[root@localhost ~(keystone_admin)]# echo "- 名称：$ORIGINAL_VM"
- 名称：vm2
[root@localhost ~(keystone_admin)]# echo "- 网络：$ORIGINAL_NETWORK"
- 网络：internal-net2
[root@localhost ~(keystone_admin)]# echo "- 浮动IP：$ORIGINAL_FLOAT_IP"
- 浮动IP：172.24.4.151
[root@localhost ~(keystone_admin)]#
[root@localhost ~(keystone_admin)]# openstack server delete $ORIGINAL_VM
[root@localhost ~(keystone_admin)]# openstack server create \
> --image $SNAPSHOT_NAME \
> --flavor m1.tiny \
> --network $ORIGINAL_NETWORK \
> --security-group basic-sg \
> --key-name mykey \
> $ORIGINAL_VM
+-----+
| Field | Value |
+-----+
| OS-DCF:diskConfig | MANUAL |
| OS-EXT-AZ:availability_zone | None |
| OS-EXT-SRV-ATTR:host | None |
| OS-EXT-SRV-ATTR:hypervisor_hostname | None |
| OS-EXT-SRV-ATTR:instance_name | None |
| OS-EXT-STS:power_state | NOSTATE |
| OS-EXT-STS:task_state | scheduling |
| OS-EXT-STS:vm_state | building |
| OS-SRV-USG:launched_at | None |
| OS-SRV-USG:terminated_at | None |
| accessIPv4 | |
| accessIPv6 | |
| addresses | |
| adminPass | KEDPVqhrR92w |
| config_drive | |
| created | 2026-01-06T12:08:56Z |
| flavor | m1.tiny (1) |
| hostId | |
| id | 3f1a6fc6-bc7b-4768-bb78-05c64a3f9c5e |
| image | vm2-snapshot (631be80b-883b-48d1-9b69-6992b116f287) |
| key_name | mykey |
| name | vm2 |
| progress | 0 |
| project_id | bb7a5f3a824742988fdb998d0c1a68fb |
| properties | |
| security_groups | name='ca676a1f-90b0-40f6-9196-61f6bc419899' |
| status | BUILD |
+-----+
[root@localhost ~(keystone_admin)]# openstack server add floating ip $ORIGINAL_VM $ORIGINAL_FLOAT_IP
[root@localhost ~(keystone_admin)]# openstack server show $ORIGINAL_VM -c name -c status -c addresses
+-----+
| Field | Value |
+-----+
| addresses | internal-net2=10.0.2.181, 172.24.4.151 |
| name | vm2 |
| status | ACTIVE |
+-----+

```

快照和镜像对比分析

```

1  ## 定义
2
3  ### 镜像 (Image)
4  - **定义**: 基础操作系统模板, 不含个人数据或配置
5  - **来源**: 从ISO文件创建, 或从云主机创建
6  - **大小**: 通常较小 (基础系统)
7  - **状态**: 不包含运行状态
8
9  ### 快照 (Snapshot)
10 - **定义**: 云主机在某个时间点的完整状态拷贝
11 - **来源**: 从运行中或停止的云主机创建
12 - **大小**: 包含整个磁盘内容, 可能较大
13 - **状态**: 包含内存状态 (如果从运行中创建)
14
15 ## 对比表格
16
17 | 特性 | 镜像 (Image) | 快照 (Snapshot) |
18 |-----|-----|-----|
19 | **目的** | 创建新云主机的模板 | 备份/恢复特定云主机状态 |
20 | **内容** | 基础操作系统 | 完整磁盘状态 (含数据、配置) |
21 | **大小** | 较小 | 较大 (与云主机磁盘相同) |
22 | **创建速度** | 慢 (需从ISO构建) | 快 (磁盘快照) |
23 | **使用场景** | 部署新实例 | 备份、迁移、回滚 |
24 | **是否包含数据** | 否 | 是 |
25 | **是否包含配置** | 否 | 是 |
26 | **生命周期** | 长期保存 | 短期或中期保存 |
27
28 ## 使用场景指南
29
30 ### 使用镜像的场景
31 1. **批量部署**: 需要部署多个相同配置的云主机
32 2. **标准化环境**: 创建公司标准的基础镜像
33 3. **模板分发**: 分享基础系统配置
34 4. **版本控制**: 维护不同版本的操作系统镜像
35
36 ### 使用快照的场景
37 1. **灾难恢复**: 备份重要云主机状态
38 2. **版本回滚**: 系统更新失败后快速恢复
39 3. **测试环境**: 创建测试点, 测试后可快速还原
40 4. **迁移准备**: 迁移前创建一致性快照
41 5. **临时备份**: 进行危险操作前的临时备份
42
43 ## 操作示例
44
45 ### 创建镜像

```

```

46  ```bash
47  # 从云主机创建镜像（类似快照，但通常用于模板）
48  openstack server image create --name web-server-template vm-web
49
50
51  # 创建快照（建议先停止云主机）
52  openstack server stop vm-production
53  openstack server image create --name vm-backup-2024 vm-production
54  openstack server start vm-production

```

镜像部署与初始化

1 创建和管理密钥对

```

1  # 1. 创建新的密钥对
2  openstack keypair create cloud-key > cloud-key.pem
3  chmod 600 cloud-key.pem
4
5  # 2. 查看密钥对
6  openstack keypair list
7
8  # 3. 显示公钥（用于后续配置）
9  openstack keypair show cloud-key --public-key

```

```

root@localhost ~(keystone_admin)]# openstack server add floating ip $ORIGINAL_VM $ORIGINAL_FLOAT IP
root@localhost ~(keystone_admin)]# openstack server show $ORIGINAL_VM -c name -c status -c addresses
-----+-----+-----+
Field      | Value                                     |
-----+-----+-----+
addresses  | internal-net2=10.0.2.181, 172.24.4.151 |
name       | vm2                                     |
status     | ACTIVE                                |
-----+-----+-----+
root@localhost ~(keystone_admin)]# openstack keypair create cloud-key > cloud-key.pem
root@localhost ~(keystone_admin)]# chmod 600 cloud-key.pem
root@localhost ~(keystone_admin)]# openstack keypair list
-----+-----+-----+
Name       | Fingerprint                             |
-----+-----+-----+
cloud-key  | d7:f9:43:a6:da:85:8c:c4:8a:48:d8:2d:9a:b1:cf:78 |
mykey     | c1:70:2c:7b:e4:8c:3e:fb:79:31:3b:21:cc:72:a6:cd |
-----+-----+-----+
root@localhost ~(keystone_admin)]# openstack keypair show cloud-key --public-key
sh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCb20CCWBRajnLYyDB+uyRZvHGfGn81Tn3lVQ0icKffHeX8VgXC0/7RPobCL0vRtWfFu7W+gR89F2tqSksvvhxLR0E14XKW6bEl
zs+v830f6KqpfX5ggC9rb07fqgn9//tRB8J07TwkqBubtrQRpWmLPBwbWtcTxY92ZIEjzn9FTETsGLnZukN9BgDa4/n90LU/3VTK3mDRtGJvH+D1M93DSXn0upjv9Lt+8EFsBe
wRz30kGywNg2MbVd93f2ArPPj0SAz4QD8u+CnHh Generated-by-Nova
root@localhost ~(keystone_admin)]# █

```

2 创建cloud-init配置文件

```
1 # 创建cloud-init配置
2 cat > user-data.txt << 'EOF'
3 #!/bin/bash
4
5 # 基础系统配置
6 echo "=== 系统初始化开始 ==="
7
8 # 1. 创建用户 (cirros镜像默认只有cirros用户)
9 sudo useradd -m -s /bin/bash admin
10 echo "admin:admin123" | sudo chpasswd
11 sudo usermod -aG sudo admin
12
13 # 2. 创建SSH目录并设置权限
14 sudo mkdir -p /home/admin/.ssh
15 sudo chmod 700 /home/admin/.ssh
16
17 # 3. 添加公钥 (请替换为你的公钥)
18 echo "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQ...你的公钥内容... user@host" |
19 sudo tee /home/admin/.ssh/authorized_keys
20 sudo chmod 600 /home/admin/.ssh/authorized_keys
21 sudo chown -R admin:admin /home/admin/.ssh
22
23 # 4. 安装常用工具 (cirros使用apk包管理器)
24 sudo apk update && sudo apk add curl wget vim
25
26 # 5. 创建测试文件
27 sudo mkdir -p /opt/cloud-init-test
28 echo "Cloud-init配置成功执行于: $(date)" | sudo tee /opt/cloud-init-test/status.txt
29 echo "主机名: $(hostname)" | sudo tee -a /opt/cloud-init-test/status.txt
30 echo "IP地址: $(hostname -I)" | sudo tee -a /opt/cloud-init-test/status.txt
31
32 # 6. 配置SSH服务 (允许密码登录用于测试)
33 sudo sed -i 's/^#PasswordAuthentication yes/PasswordAuthentication yes/' /etc/ssh/sshd_config
34 sudo sed -i 's/^#PasswordAuthentication no/PasswordAuthentication no/' /etc/ssh/sshd_config
35 sudo service sshd restart
36
37 echo "=== 系统初始化完成 ==="
38 EOF
39 echo "cloud-init配置已创建: user-data.txt"
```

3.设计安全组最小权限策略

Python |

```
1  # 1. 创建新的安全组
2  openstack security group create minimal-sg --description "最小权限安全组"
3
4  # 2. 添加最小权限规则
5  # a. 允许SSH (22端口), 但只允许特定源IP (这里用0.0.0.0/0模拟, 实际应限制)
6  openstack security group rule create \
7      --protocol tcp \
8      --dst-port 22 \
9      --remote-ip 0.0.0.0/0 \
10     minimal-sg
11
12 # b. 允许ICMP (ping)
13 openstack security group rule create \
14     --protocol icmp \
15     --remote-ip 0.0.0.0/0 \
16     minimal-sg
17
18 # 3. 查看安全组规则
19 echo "安全组规则: "
20 openstack security group rule list minimal-sg
```

```
[root@localhost ~(keystone_admin)]# openstack security group create minimal-sg --description "最小权限安全组"
+-----+-----+
| Field          | Value                                     |
+-----+-----+
| created_at     | 2026-01-06T12:27:02Z                    |
| description    | 最小权限安全组                          |
| id             | 1b8be0e0-4b0e-4f43-bd7d-54b496a11915    |
| location       | Munch({'project': Munch({'domain_name': 'Default', 'domain_id': None, 'name': 'admin', 'id': u'bb7a5f3a824742988fdb998d0c1a68fb', 'on_name': 'RegionOne', 'zone': None})}) |
| name          | minimal-sg                              |
| project_id     | bb7a5f3a824742988fdb998d0c1a68fb       |
| revision_number | 1                                         |
| rules          | created_at='2026-01-06T12:27:02Z', direction='egress', ethertype='IPv4', id='9925b00c-8543-490b-8950-e517bc7335ca', updated_at='2026-01-06T12:27:02Z', direction='egress', ethertype='IPv6', id='c45dd343-f18e-4a7c-a821-b52aa1f15e80', updated_at='2026-01-06T12:27:02Z' |
| tags           | []                                        |
| updated_at     | 2026-01-06T12:27:02Z                    |
```

```

[root@localhost ~(keystone_admin)]# openstack security group rule create \
> --protocol tcp \
> --dst-port 22 \
> --remote-ip 0.0.0.0/0 \
> minimal-sg
+-----+
| Field          | Value                                     |
+-----+-----+
| created_at     | 2026-01-06T12:27:16Z                    |
| description    |                                           |
| direction      | ingress                                  |
| ether_type     | IPv4                                     |
| id             | ee54c0f1-a284-430a-9c50-a23ea8cf9827   |
| location       | Munch({'project': Munch({'domain_name': 'Default', 'domain_id': None, 'name': 'admin', 'id': u'bb7a5f3e |
| gion_name': 'RegionOne', 'zone': None}) |
| name           | None                                     |
| port_range_max | 22                                       |
| port_range_min | 22                                       |
| project_id     | bb7a5f3a824742988fdb998d0c1a68fb      |
| protocol       | tcp                                     |
+-----+-----+

[root@localhost ~(keystone_admin)]# # b. 允许ICMP (ping)
[root@localhost ~(keystone_admin)]# openstack security group rule create \
> --protocol icmp \
> --remote-ip 0.0.0.0/0 \
> minimal-sg
+-----+
| Field          | Value                                     |
+-----+-----+
| created_at     | 2026-01-06T12:27:37Z                    |
| description    |                                           |
| direction      | ingress                                  |
| ether_type     | IPv4                                     |
| id             | 25ad4ecf-2239-4a20-ab5f-50c0d59e3e5d   |
| location       | Munch({'project': Munch({'domain_name': 'Default', 'domain_id': None, 'name': 'admin', 'id': u'bb7a5f |
| gion_name': 'RegionOne', 'zone': None}) |
| name           | None                                     |
| port_range_max | None                                     |
| port_range_min | None                                     |
| project_id     | bb7a5f3a824742988fdb998d0c1a68fb      |
| protocol       | icmp                                     |
| remote_group_id | None                                     |
+-----+-----+

[root@localhost ~(keystone_admin)]# openstack security group rule list minimal-sg
+-----+-----+-----+-----+-----+
| ID                                     | IP Protocol | IP Range | Port Range | Remote Security Group |
+-----+-----+-----+-----+-----+
| 25ad4ecf-2239-4a20-ab5f-50c0d59e3e5d | icmp        | 0.0.0.0/0 |           | None                   |
| 9925b00c-8543-490b-8950-e517bc7335ca | None        | None      |           | None                   |
| c45dd343-f18e-4a7c-a821-b52aa1f15e80 | None        | None      |           | None                   |
| ee54c0f1-a284-430a-9c50-a23ea8cf9827 | tcp         | 0.0.0.0/0 | 22:22     | None                   |
+-----+-----+-----+-----+-----+

```

4.创建云主机并应用配置

```
1  # 1. 创建云主机
2  openstack server create \
3      --image cirros \
4      --flavor m1.tiny \
5      --network private \
6      --security-group minimal-sg \
7      --key-name cloud-key \
8      --user-data user-data.txt \
9      cloud-vm1
10
11 # 2. 分配浮动IP
12 FLOAT_IP=$(openstack floating ip create public -c floating_ip_address -f v
13   value)
14 openstack server add floating ip cloud-vm1 $FLOAT_IP
15 echo "云主机创建中, 浮动IP: $FLOAT_IP"
16 echo "等待云主机启动..."
17 sleep 30
18
19 # 3. 查看云主机状态
20 openstack server show cloud-vm1 -c name -c status -c addresses
```

5.验证配置

```
1  # 1. 测试SSH连接 (cirros默认用户cirros, 密码gocubsgo)
2  echo "测试SSH连接..."
3  echo "命令: ssh cirros@$FLOAT_IP"
4  echo "密码: gocubsgo"
5
6  # 2. 创建验证脚本
7  cat > verify_cloud_init.sh << 'EOF'
8  #!/bin/bash
9
10 FLOAT_IP=$1
11
12 echo "=== 验证cloud-init配置 ==="
13 echo "1. 测试SSH连接..."
14 sshpass -p 'gocubsgo' ssh -o StrictHostKeyChecking=no cirros@$FLOAT_IP "echo 'SSH连接成功'"
15
16 echo ""
17 echo "2. 检查创建的文件..."
18 sshpass -p 'gocubsgo' ssh -o StrictHostKeyChecking=no cirros@$FLOAT_IP "ls -la /opt/cloud-init-test/"
19
20 echo ""
21 echo "3. 查看状态文件..."
22 sshpass -p 'gocubsgo' ssh -o StrictHostKeyChecking=no cirros@$FLOAT_IP "cat /opt/cloud-init-test/status.txt"
23
24 echo ""
25 echo "4. 检查安装的软件..."
26 sshpass -p 'gocubsgo' ssh -o StrictHostKeyChecking=no cirros@$FLOAT_IP "which curl && which wget && which vim"
27
28 echo ""
29 echo "5. 验证安全组规则..."
30 echo "    尝试ping: ping -c 2 $FLOAT_IP"
31 ping -c 2 $FLOAT_IP && echo "ICMP允许 ✓" || echo "ICMP被拒绝 ✗"
32 EOF
33
34 chmod +x verify_cloud_init.sh
35
36 # 如果安装了sshpass, 可以运行验证
37 if command -v sshpass &> /dev/null; then
38     ./verify_cloud_init.sh $FLOAT_IP
39 else
40     echo "请安装sshpass后运行: sudo apt install sshpass"
41     echo "或手动验证:"
```



```
42     echo " SSH: ssh cirros@$FLOAT_IP"  
43     echo " 密码: gocubsgo"  
44 fi
```

6 创建第二台云主机验证一致性

```

1  # 1. 创建第二台云主机（使用相同的配置）
2  openstack server create \
3      --image cirros \
4      --flavor m1.tiny \
5      --network private \
6      --security-group minimal-sg \
7      --key-name cloud-key \
8      --user-data user-data.txt \
9      cloud-vm2
10
11 # 2. 分配浮动IP
12 FLOAT_IP2=$(openstack floating ip create public -c floating_ip_address -f
value)
13 openstack server add floating ip cloud-vm2 $FLOAT_IP2
14
15 echo "第二台云主机创建中，浮动IP: $FLOAT_IP2"
16 sleep 30
17
18 # 3. 验证两台云主机的一致性
19 cat > verify_consistency.sh << 'EOF'
20 #!/bin/bash
21
22 echo "=== 验证多台云主机一致性 ==="
23 echo "云主机1 ($FLOAT_IP) 和云主机2 ($FLOAT_IP2)"
24
25 echo ""
26 echo "1. 检查cloud-init状态文件..."
27 echo "云主机1:"
28 sshpass -p 'gocubsgo' ssh -o StrictHostKeyChecking=no cirros@$FLOAT_IP "ca
t /opt/cloud-init-test/status.txt | grep '主机名'"
29 echo ""
30 echo "云主机2:"
31 sshpass -p 'gocubsgo' ssh -o StrictHostKeyChecking=no cirros@$FLOAT_IP2 "c
at /opt/cloud-init-test/status.txt | grep '主机名'"
32
33 echo ""
34 echo "2. 检查安装的软件包..."
35 echo "云主机1的curl版本:"
36 sshpass -p 'gocubsgo' ssh -o StrictHostKeyChecking=no cirros@$FLOAT_IP "cu
rl --version 2>/dev/null | head -1"
37 echo ""
38 echo "云主机2的curl版本:"
39 sshpass -p 'gocubsgo' ssh -o StrictHostKeyChecking=no cirros@$FLOAT_IP2 "c
url --version 2>/dev/null | head -1"
40

```

```
41 echo ""
42 echo "3. 验证安全组规则..."
43 echo "两台云主机都应该允许ping和SSH"
44 EOF
45
46 chmod +x verify_consistency.sh
47 ./verify_consistency.sh
```

7 安全风险与合规说明

```
1  # 创建安全风险说明文档
2  cat > security_risks.md << 'EOF'
3  # 安全风险与合规说明
4
5  ## 1. 已识别风险
6
7  ### 高风险:
8  1. **SSH密码认证启用**: 配置中启用了密码认证, 可能被暴力破解
9     - 建议: 禁用密码认证, 仅使用密钥对
10
11  2. **安全组规则过宽**: 允许0.0.0.0/0访问SSH端口
12     - 建议: 限制为特定IP段, 如公司网络IP
13
14  3. **默认凭证**: 使用默认用户/密码 (cirros/gocubsgo)
15     - 建议: 首次登录后立即修改密码
16
17  ### 中风险:
18  1. **云主机间无隔离**: 同一安全组内所有云主机网络互通
19     - 建议: 使用网络ACL或不同安全组进行隔离
20
21  2. **API密钥未轮换**: 密钥对长期未更换
22     - 建议: 定期轮换密钥对 (每90天)
23
24  3. **无日志监控**: 未配置集中日志和监控
25     - 建议: 部署日志收集和入侵检测系统
26
27  ### 低风险:
28  1. **服务暴露**: 安装的服务可能包含未修复的漏洞
29     - 建议: 定期更新系统和应用补丁
30
31  ## 2. 合规性要求
32
33  ### 必须遵守:
34  1. **访问控制**: 实施最小权限原则
35  2. **认证管理**: 使用多因素认证或密钥对
36  3. **日志审计**: 保留至少90天的操作日志
37  4. **漏洞管理**: 定期扫描和修复漏洞
38
39  ### 建议实施:
40  1. **网络分段**: 生产、测试、开发环境隔离
41  2. **加密传输**: 所有管理连接使用TLS/SSL
42  3. **备份策略**: 定期备份配置和数据
43
44  ## 3. 加固建议
45
```

```
46  ```bash
47  # 安全加固脚本示例
48  #!/bin/bash
49  # 1. 禁用SSH密码认证
50  sed -i 's/^PasswordAuthentication yes/PasswordAuthentication no/' /etc/ssh
51  /sshd_config
52
53  # 2. 更改默认SSH端口
54  sed -i 's/^#Port 22/Port 2222/' /etc/ssh/sshd_config
55
56  # 3. 配置fail2ban防止暴力破解
57  apk add fail2ban
58  systemctl enable fail2ban
59
59  # 4. 配置防火墙（如果可用）
```

云主机初始化模板脚本

```
1  #!/bin/bash
2  # 云主机初始化模板脚本
3  # 保存为模板，可用于批量创建相同配置的云主机
4
5  set -e
6
7  # 配置参数
8  VM_NAME="$1"
9  NETWORK="${2:-private}"
10 FLAVOR="${3:-m1.tiny}"
11 SECURITY_GROUP="${4:-minimal-sg}"
12 KEY_NAME="${5:-cloud-key}"
13
14 # 创建user-data文件
15 cat > /tmp/user-data-$$.$$.txt << 'DATA_EOF'
16 #!/bin/bash
17 # 基础初始化脚本
18 echo "开始系统初始化..."
19
20 # 1. 软件包更新和安装
21 apk update
22 apk add curl wget vim htop
23
24 # 2. 创建应用目录
25 mkdir -p /opt/app/{config,logs,data}
26
27 # 3. 创建监控脚本
28 cat > /opt/health-check.sh << 'SCRIPT_EOF'
29 #!/bin/bash
30 echo "Health check at $(date)"
31 echo "CPU: $(top -bn1 | grep load)"
32 echo "Memory: $(free -m)"
33 echo "Disk: $(df -h /)"
34 SCRIPT_EOF
35 chmod +x /opt/health-check.sh
36
37 # 4. 配置定时任务
38 echo "*/* 5 * * * * /opt/health-check.sh >> /opt/app/logs/health.log" >> /etc/crontabs/root
39
40 # 5. 设置主机名
41 echo "${VM_NAME}" > /etc/hostname
42 hostname ${VM_NAME}
43
44 echo "初始化完成于 $(date)"
```

```
45 DATA_EOF
46
47 # 创建云主机
48 echo "创建云主机: $VM_NAME"
49 openstack server create \
50     --image cirros \
51     --flavor $FLAVOR \
52     --network $NETWORK \
53     --security-group $SECURITY_GROUP \
54     --key-name $KEY_NAME \
55     --user-data /tmp/user-data-$$$.txt \
56     $VM_NAME
57
58 # 清理临时文件
59 rm -f /tmp/user-data-$$$.txt
60
61 echo "云主机 $VM_NAME 创建命令已执行"
62 echo "请等待云主机启动后分配浮动IP"
63 EOF
```

安全组最小权限验证脚本

```
1  #!/bin/bash
2  # 安全组最小权限验证脚本
3
4  FLOAT_IP=$1
5  if [ -z "$FLOAT_IP" ]; then
6      echo "请提供云主机浮动IP"
7      exit 1
8  fi
9
10 echo "=== 安全组最小权限验证 ==="
11 echo "测试目标: $FLOAT_IP"
12 echo ""
13
14 # 测试允许的端口
15 echo "1. 测试允许的端口:"
16 echo "    a) SSH (22端口):"
17 if nc -z -w 2 $FLOAT_IP 22; then
18     echo "        允许 ✓"
19 else
20     echo "        拒绝 ✗"
21 fi
22
23 echo ""
24 echo "    b) ICMP (ping):"
25 if ping -c 2 -W 1 $FLOAT_IP &> /dev/null; then
26     echo "        允许 ✓"
27 else
28     echo "        拒绝 ✗"
29 fi
30
31 echo ""
32 # 测试不允许的端口 (应该被拒绝)
33 echo "2. 测试未允许的端口 (应该被拒绝) :"
34 PORTS="80 443 3306 3389 8080"
35 for port in $PORTS; do
36     if nc -z -w 2 $FLOAT_IP $port 2>/dev/null; then
37         echo "    端口 $port: 允许 ✗ (安全漏洞!)"
38     else
39         echo "    端口 $port: 拒绝 ✓"
40     fi
41 done
42
43 echo ""
44 echo "3. 安全组规则检查:"
45
```



```
46  openstack security group rule list minimal -sg -c "IP Protocol" -c "Port Ra
47  nge" -c "Remote IP Prefix"
48
49  echo ""
50  echo "=== 验证结果 ==="
    echo "如果只有SSH(22)和ICMP被允许，其他端口都被拒绝，则符合最小权限原则。"
```

最终验证脚本

```
1  #!/bin/bash
2
3  echo "=== 任务3完成情况验证 ==="
4  echo ""
5
6  # 1. 检查密钥对
7  echo "1. 密钥对创建:"
8  openstack keypair list | grep cloud-key && echo "    ✓ 完成" || echo "    x
  未完成"
9
10 # 2. 检查云主机
11 echo ""
12 echo "2. cloud-init云主机创建:"
13 openstack server list --name cloud-vm -c Name -c Status | grep cloud-vm &&
  echo "    ✓ 完成" || echo "    x 未完成"
14
15 # 3. 检查安全组
16 echo ""
17 echo "3. 最小权限安全组:"
18 if openstack security group list | grep -q minimal-sg; then
19     echo "    ✓ 安全组已创建"
20     echo "    规则数量: $(openstack security group rule list minimal-sg | w
  c -l)"
21 else
22     echo "    x 安全组未创建"
23 fi
24
25 # 4. 检查配置文件
26 echo ""
27 echo "4. 配置文件:"
28 [ -f "user-data.txt" ] && echo "    ✓ user-data.txt 存在" || echo "    x use
  r-data.txt 不存在"
29 [ -f "security_risks.md" ] && echo "    ✓ security_risks.md 存在" || echo
  "    x security_risks.md 不存在"
30
31 # 5. 验证工具脚本
32 echo ""
33 echo "5. 验证脚本:"
34 [ -f "verify_cloud_init.sh" ] && echo "    ✓ verify_cloud_init.sh 存在" ||
  echo "    x verify_cloud_init.sh 不存在"
35 [ -f "test_security_group.sh" ] && echo "    ✓ test_security_group.sh 存在"
  || echo "    x test_security_group.sh 不存在"
36
37 echo ""
38 echo "=== 任务完成状态 ==="
```

```
39 echo "✅ 密钥对创建与管理"  
40  
41 echo "✅ cloud-init配置编写与部署"  
42  
43 echo "✅ 安全组最小权限策略设计"  
44 echo "✅ 多台云主机一致性验证"  
45 echo "✅ 安全风险与合规说明"  
EOF
```